

Very low n-3 long-chain polyunsaturated fatty acid status in Austrian vegetarians and vegans.

BACKGROUND/AIMS: The objective of the study was to collect data on dietary fat intake of omnivores, vegetarians, vegans and semi-omnivores as well as its impact on n-3 and n-6 fatty acids in long-term markers such as sphingolipids, phosphatidylcholine (PC), phosphatidylserine (PS), phosphatidylethanolamine (PE) as well as the calculated sphingo- and phospholipids (SPL) of erythrocytes. **METHOD:** The present observational study included 98 Austrian adult volunteers of both genders, of which 23 were omnivores, 25 vegetarians, 37 vegans, and 13 semi-omnivores. Information on anthropometry using measured body weight and height was obtained. The amount and composition of ingested fat were calculated from 24-hour recalls and the fatty acid pattern in the phospholipids was assessed using gas chromatography. **RESULTS:** The unbalanced n-6/n-3 ratio and the limited dietary sources of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) in vegans and vegetarians led to reductions in C20:5n-3, C22:5n-3, C22:6n-3 and total n-3 fatty acids in SPL, PC, PS and PE compared with omnivores and semi-omnivores. The total content of polyunsaturated fatty acids, monounsaturated fatty acids and saturated fatty acids remained unchanged. **CONCLUSION:** The vegetarian diet, with an average n-6/n-3 ratio of 10/1, promotes biochemical n-3 tissue decline. To ensure physical, mental and neurological health vegetarians have to reduce the n-6/n-3 ratio with an additional intake of direct sources of EPA and DHA, regardless of age and gender. (c) 2008 S. Karger AG, Basel.